An Intelligent Marathi Sign Language Recognition Framework Using Spatial Features and CNN

Atharva G Zope
Computer Engineering(Regional Langage)
Pimpri Chinchwad College of Engineering
Pune, India
atharva.zope23@pccoepune.org

Adwait A.Kamble
Computer Engineering(Regional Language)
Pimpri Chinchwad College of Engineering
Pune, India
adwait.kamble23@pccoepune.org

Prof. Anandkumar Birajdar
Computer Engineering(Regional Language)
Pimpri Chinchwad College of Engineering
Pune, India
anand.birajdar@pccoepune.org

Yadnesh Patil
Computer Engineering(Regional Language)
Pimpri Chinchwad College of Engineering
Pune, India
yadnesh.patil23@pccoepune.org

Abhay Bormale
Computer Engineering(Regional Language)
Pimpri Chinchwad College of Engineering
Pune, India
abhay.bormale23@pccoepune.org

***Abstract*—The hearing-impaired community gets big hassles when they attempt to speak with people who do not have hearing problems, because the hearing system does not really work well for Marathi, which is basically a local language. Existing sign language recognition systems usually rely on fixed word-to-word dictionary type translations, and because of that they cannot really build sentences that vary as situations change. In this research we present a full setup that lets users recognize Marathi Sign Language using multiple approaches while also translating into different languages. The Advanced Image and Video Processing (IVP) part helps us reduce both environmental noise and skin-tone bias that show up in the original video streams. We also built a visual pipeline that blends Region of Interest (ROI) extraction with Gaussian smoothing , and Canny edge detection, so we can isolate spatio temporal topological contours of hand gestures more precisely. After that, the system captures structural features, these are then treated as Marathi root characters, before they are passed to a natural language processing (NLP) engine, which runs at high speed through a large language model (LLM) system. This hybrid system is meant to support users so they can "hear" phonetic style sounds while it generates properly spoken Marathi sentences, which can be used for basic language translation. The outcomes of the research indicated that the method which tries to detect mathematical hand geometry before classification gives better computation scores, and improves overall system strength. The proposed system reached a solid overall testing accuracy of 91.6%. The combination of structural edge detection and generative AI creates a system that enables fast and efficient communication that helps deaf people access services while maintaining safety through SOS emergency systems.**

Keywords— Marathi Sign Language (MSL), Image Processing, Canny Edge Detection, Gaussian Smoothing, ROI Extraction, Natural Language Processing, Deep Learning, Accessibility, Marathi.

I. INTRODUCTION

Human-Computer Interaction (HCI) has progressed from its original text-based input methods to its current development of gesture-based interfaces which provide users with basic nonverbal interface solutions. The society demands systems which enable access to technology for all users including people who have disabilities as technology becomes more integrated into their everyday activities. The domain faces its biggest obstacle because hearing-impaired people cannot effectively communicate with other people. The development of strong vision systems needs to interpret human gestures requires small systems which can recognize human movements. [8] The combination of Digital Image Processing methods with Computer Vision techniques enables us to create mathematical models which interpret hand gestures through their visual signals to produce digital text and speech output.[4] Through verbal communication people can share their thoughts and feelings and their knowledge with one another. Most people use spoken language as their main way of communicating with others. The hearing and speech-impaired community uses Sign Language as their main method of communication. The system functions as a visual language through its use of hand gestures and orientation and movement together with facial expressions to create meaningful content. This research explores the large communication barriers that persons with hearing and speech impairments have to deal within the state of Maharashtra, which is significantly based upon their use of Marathi Sign Language (MSL) as their primary means of communication. Although considerable levels of research are available for American Sign Language (ASL) and overall Indian Sign Language (ISL), there is a distinct void in the availability of any credible and developed real-time recognition systems that have been specifically created to recognize the Marathi Varnmala (gestures forming the Devanagari script). Further, solutions created to provide support to the community typically do not incorporate the syntactic nuances required to effectively communicate regionally signified languages, thus creating a digital divide and minimising accessibility. [1] Advances in Deep Learning have greatly changed the field of computer vision but using only Convolutional Neural Networks (CNNs) creates many problems for their use in real-time applications. An approach that does not use any features (pure AI) by directly sending a neural network a raw RGB image creates a large amount of computational complexity and duplicate data. The network is then required to learn both the gestures of the user's hands while also filtering out the excess noise created by the background, different light conditions, and different camera/noise characteristics. The process of learning features requires an extremely large amount of input data as well as a significant amount of GPU processing power; therefore, it is not usually possible to process the data using a standard desktop computer. [1]

The research presented here is motivated by the increased efficiency of using a Hybrid Approach, which combines classical Spatial Domain Image Processing Techniques (such as Thresholding, Morphological Operations and Edges) with modern Deep Learning Techniques. Balancing the progress in Artificial Intelligence and Machine Learning (AI/ML), this hybrid setup kind of shows how disciplined algorithmic pre-processing can make AI more accessible. In a more formal sense, it mathematically shrinks the input dimension s first, before it even reaches the neural network part. With that, the system offers fast low-latency AI inference on edge devices, so there is no need for cloud computing, while classification accuracy still stays high.[10]

Table 1 Comparative Analysis of Pure Deep Learning vs. Proposed Hybrid Approach.

Parameter	Pure Deep Learning (Standard CNN)	Proposed Hybrid System (IVP + CNN)
Input Data	Raw RGB Pixels (High Dimensionality).	Processed Binary/Edge Mask (Low Dimensionality).
Computational Load	High; requires learning invariant features from scratch.	**Low**; Pre-processing handles noise and background removal.
Training Time	Long; slower convergence due to complex input.	**Fast**; Network focuses only on shape classification.
Robustness	Sensitive to background clutter and lighting.	**High**; Thresholding & Edge Detection filter out environmental noise.
Syllabus Concept	Automated Feature Learning.	Spatial Filtering & Gradient Feature Extraction.

The aim of this paragraph is to describe the parameters (or limits) of the research as defined by the construction of a new, non-intrusive, vision-based (HCI) system. Unlike the current method of using sensors, such as data gloves or other similar devices to mediate human-computer interaction (like with the current camera), this research system will develop a complete human-computer interaction system solely with digital image acquisition via a standard webcam, within the current research methodology framework. Specifically, the scope is focused on the ability to recognize the Marathi Varnmala (which consists of vowels and consonants) in addition to the limited number of common, static phrases that the user may use when communicating on a daily basis. [5]

This research has the following primary objectives designed to systematically address the challenges associated with image processing and pattern recognition.

Implement Robust Digital Image Acquisition: Developing a module to capture the video in real-time and perform any necessary sampling and quantization to change existing visual analog data into discrete digital video frames, which then can be used for processing. [2]

Apply Spatial Domain Enhancement: Using procedures such as smoothing spatial filters (e.g. Gaussian Blur) for noise reduction and morphological operations (i.e. Erosion and Dilation) on the binary mask of the hand to improve it. [3]

Perform Accurate Segmentation: Using algorithmic thresholding techniques (i.e Otsu's method or Adaptive Thresholding) to accurately separate the hand's foreground from its complex background. [5]

To Extract Structural Features: Using gradient-based filters (i.e Canny Edge Detection) to produce the geometric shape and define the boundary features of the hand with minimal dependency on colour information. [12]

To Develop a Hybrid Classification Model: Train a Convolutional Neural Network (CNN) on the edge maps that were pre-processed for accurate classification of Marathi characters.

Enable Audio-Visual Feedback: The integration of a Text-To-Speech (TTS) engine to provide the end-user with a vocally represented prediction of the character, thereby completing the end-user's communications loop. [1]

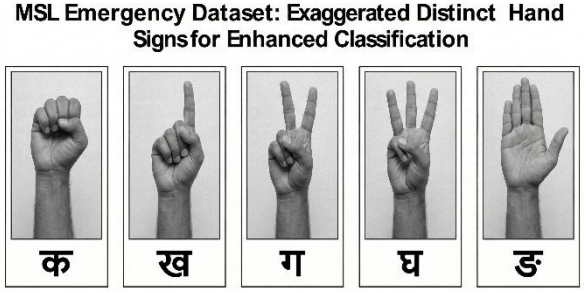


Figure 1: Scope of Dataset - Selected Marathi Varnmala (Vowels and Consonants).

This paper brings novel contributions for regional sign language translation, aiming to close some gaps that were sort of left open before. In practice the whole thing is broken into three main parts, but not like, rigidly.

- First, there's an Edge-Viable Hybrid Architecture: a highly optimized pipeline that links spatial domain filtering (HSV , Morphological filter's , Canny Edge's) with a lighter CNN. This combination cuts down the parameter count a lot, and still manages real-time 30 FPS inference, and you don't need any cloud dependency for it.
- Second, the Curated MSL Dataset: a robust 48-class set built for illumination invariance. It's a Marathi Varnmala dataset, and it includes more than 51,000 augmented frames gathered across multiple lighting conditions.
- Third, LLM-Driven Continuous Translation: this is where discrete CNN character recognition gets connected with Generative AI (LLMs). The goal is to generate continuous, context-aware, and grammatically correct Marathi sentences, while also keeping real-time audio synthesis in the loop.

II. LITERATURE SURVEY

Research in Sign Language Recognition (SLR) has progressed dramatically in the last decade, moving from identifying simple static gestures to interpreting complex continuous sentences. Many research efforts have investigated different approaches to SLR based upon their use of either hardware (i.e. sensor-based technologies) or software (i.e. vision-based technologies). This section of this report will discuss the latest

research and methodologies used for SLR, comparing the benefits and disadvantages of various methodologies currently in use. The section will focus on how traditional hand-crafted feature extraction methods (e.g. using spatial domain filters and edge detection) have evolved through different types of Deep Learning architectures. In addition, we will review the existing research on Indian Sign Language (ISL) and identify areas where additional research needs to be conducted for regional languages, such as Marathi, which will assist in the development of the proposed hybrid system. [4]. In the system Sign language recognition systems can generally be divided into two main categories – sensor-based and vision-based systems. [11]

Table 2 Comparison of Sensor-based and Vision-based Recognition Systems.

Feature	**Sensor-Based Systems (Data Gloves)**	**Vision-Based Systems (Proposed Approach)**
Input Modality	Direct electromechanical signals (Flex/Bend).	**Digital Image Acquisition** (2D Pixel Arrays).
User Comfort	Low; requires wearing physical hardware.	**High**; non-intrusive (contactless).
Cost	High (expensive sensors required).	**Low** (uses standard webcam).
Complexity	Hardware-intensive; simple data processing.	**Software-intensive**; requires **Image Processing** & **AI**.
Environmental Sensitivity	Low; works in dark/noisy environments.	**High**; sensitive to lighting and background clutter.
Syllabus Relevance	Signal Processing.	Digital Image Processing & Computer Vision.

As opposed to hardware-based systems, vision-based systems employ conventional cameras to digitally acquire images in an unobtrusive and lower-cost manner. The method treats the gesture detection problem as a computer vision problem where the camera is acquiring a continuous flow of 2D image data. The challenge of the vision-based system has shifted from one that is hardware based to one that is algorithmically complex, requiring high-performance image processing methodologies to extract features from pixel (image) data. [6]

With acceptable accuracy for continuous sentence recognition, most of the world has adequately studied American Sign Language (ASL). Indian Sign Language (ISL) has not been studied to the same extent; The body of work on ISL is disjointed; In contrast to Western sign languages, ISL is a distinct language with different vocabulary and grammar than English/Hindi. [2] There exists a significant void in research in regards to Mahrati Sign Language (MSL), in that there are no resources to support this language within a dataset or recognition models for use with the written form of Mahrati. Although Marathi has over 83 million speakers, the number of existing ISL systems is low, and therefore none of these really do consider the unique syntactical features and/or the entire character set of the Devanagari script associated with Mahrati. In addition, almost all of the proposed systems are designed for controlled illumination conditions, thus being ineffective when used with the typical amount of ambient noise and dynamic range produced with standard digital cameras. Through the development of a practical and effective pre-processing pipeline for MSL hand signs, this research will provide solutions to limit these locations of inaccuracy in prior research. [1] The field of Sign Language Recognition research has achieved major advancements during the past ten years which enable researchers to decode ent ire sentences instead of basic static signs. The authors of [1] paper established a method for recognizing Marathi sign language which uses Canny edge detection and image segmentation as its initial step toward sensor-free classification. The primary base document we used to test Canny's algorithm for Devanagari script authentication showed an absence of functional Text-to-Speech auditory output system which enables users to interact through spoken word because it lacked this feature. The research paper in [2] establishes a mathematical framework which serves as a foundational system for detecting Indian Sign Language through instantaneous translation. The authors established a sequence which combined grayscale conversion with binary thresholding and Canny edge detection to produce input data for the CNN system which supports efficient edge computing processing. The researchers of paper's studied how to use Convolutional Neural Networks (CNNs) with their specially developed Marathi sign language dataset. The study demonstrated that CNN architecture achieves high accuracy results; however, the research process depended too much on deep learning because it did not implement adequate spatial data processing techniques which resulted in increased system resource demands. The authors of the research paper introduced a method for skin colour detection which uses deep learning-based segmentation to separate hand movements from their surrounding environment. The system uses skin colour detection through HSV method to find ROIs which it sends to the neural network for processing. The research paper in [5] introduced a combined deep learning architecture which uses CNN and BiLSTM for immediate sign language interpretation. The design of our lightweight real-time CNN system followed guidance from previous work, yet BiLSTM implementation brought new challenges which required us to handle data in sequential form.[9]

III. PROPOSED METHODOLOGY

In this section, we will describe the theoretical aspects of the spatial domain filtering and feature extraction algorithms covered in the last section as they relate to the mathematical basis for the MSL system design. This section describes the implementation and architecture of the Hybrid Model. [6]

We propose a methodology to establish a real-time, compressed, digital-to-analog pipeline for the collection of raw video frames from a digital camera (or other type of camera) and through to the intelligent analysis of those frames using video analysis technologies. The proposed methodology will provide solutions for ensuring that the Hybrid Model is capable of achieving computationally efficient and noise-robust performance under a variety of noise-producing conditions in real-time.[6]

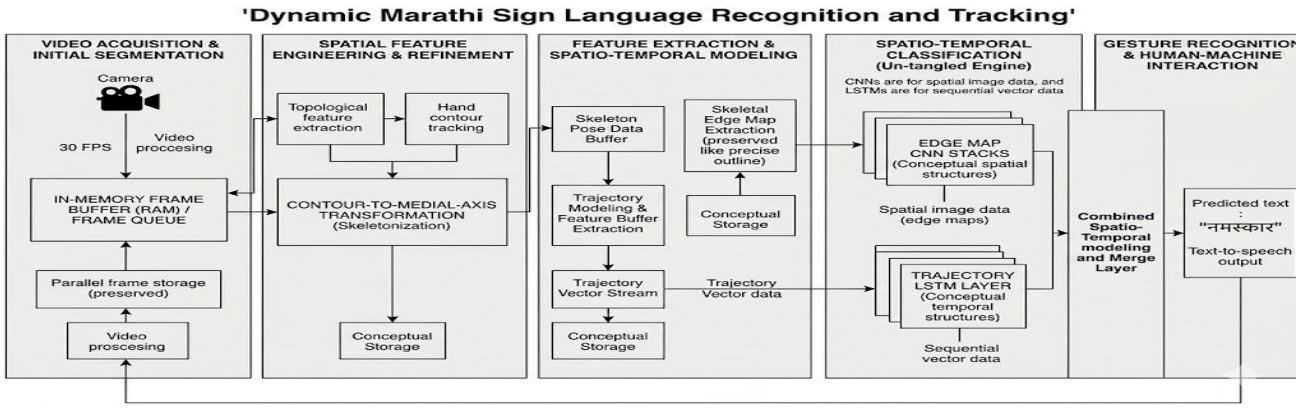


Figure 2: Proposed System Architecture Block Diagram

The designed system architecture operates as a sequential processing system which consists of multiple processing stages that enable its operation. The system accepts continuous analog video input which it processes through discrete digital components to produce both translated text and generated speech output. The system's architecture breaks down into multiple connected stages which form its high-level design structure.[4]

A good probabilistic model needs to be outstanding in discrimination and calibration:

The system begins its operation by using a standard webcam to capture active video stream. The system converts continuous analog audio signals into digital audio frames which it samples at a rate of 30 frames per second. The system processes each frame by limiting its computational work to a particular Region of Interest (ROI) which it has defined. [5]. The raw RGB frames display low resistance to the effects of illumination changes. The frames undergo conversion from RGB colour space to HSV (Hue Saturation Value) colour space to solve this problem. A Gaussian Low Pass Filter is used to eliminate high-frequency camera noise from the hand without damaging its structural integrity. [5]

This process uses binary thresholding algorithms to create a mask which separates human skin pixels from background elements in the HSV image. The system produces a binary mask which undergoes morphological operations (Opening and Closing) to eliminate single pixels of noise and complete missing shapes. [1]

The system processes refined binary masks through the Canny Edge Detection algorithm instead of using raw pixel data to train its classifier. The system uses spatial gradients to create a hand gesture outline which has a width of one pixel throughout its entire length. *[13]*

The system uses the extracted edge map which shows the basic structure of the gesture to train a Convolutional Neural Network (CNN). The CNN functions as an automatic feature extraction tool which compares existing spatial patterns with known classes of the Marathi Varnmala. [4]

The system uses the Softmax layer output which has the highest probability to determine its corresponding class. The system uses the Softmax layer output which has the highest probability to determine its corresponding class. [1]

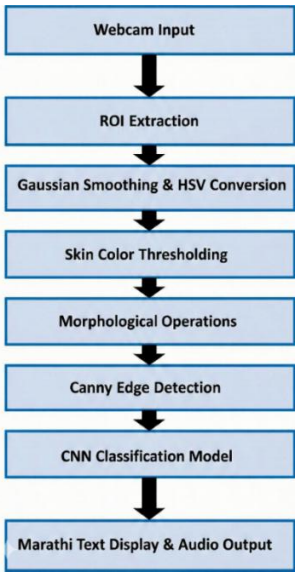


Fig 3. Proposed System Architecture Block Diagram

The development of recognition systems for regional languages encounters a major obstacle because there are no standardized datasets available. The research team aimed to create a strong custom dataset which contains complete data of Marathi Sign Language (MSL) that includes all Devanagari Varnmala vowels and consonants. The research team designed the data acquisition system to achieve proper digital image processing sampling and quantization procedures.[2]

Setup Constraints and Acquisition Parameters:

The Deep Learning model needs proper gesture representation which should be learned from basic movement patterns instead of actual environmental factors. The initial data collection process was executed under controlled conditions which maintained realistic behavior.[3]

Table 3: Creation of the Custom Marathi Dataset:

Marathi Character	English Class	Hand Gesture image	Marathi Character	English Class	Hand Gesture image
क	Ka		ख	Kha	
ग	Ga		घ	Gha	
ङ	Nga		च	Cha	
छ	Chha		ज	Ja	

The dataset contains gesture samples which demonstrate essential Marathi characters through their static form. The signer performed the gesture inside the designated area while the system automated its frame collection and processing steps for each class. The system operated without storing the unprocessed RGB image files because it conducted full spatial domain processing in real time while it stored the output binary edge maps. The architectural design requires the Convolutional Neural Network (CNN) to learn only from structural morphology information which leads to substantial decrease in input data dimensionality. For the deep learning models to satisfy those strict data requirements, the final curated dataset ended up with 51,687 augmented frames across 48 separate Marathi Varnmala classes, or basically 48 categories. We gathered everything from 12 participants, (7 Male,5 Female) with different lighting setups like Natural Daylight, Fluorescent, and Dim Indoor, and we used a standard 1080p webcam running at 30 FPS. Later, we did a split of 70% for training and 30% for validation, so the model should generalize more robustly.[6]

In order to ensure that there was an adequate supply of data for MSL deep learning models and because there was no internationally recognized standard for MSL datasets a new database was created specifically for this study. To enhance real-world robustness, each subject's motion was recorded under several different lighting conditions. In addition, in order to eliminate the risk of a CNN memorizing a specific background environment, the dataset was designed with a large number of backgrounds. The specifications of the dataset and experimental design are provided in Table 1.

Table 4: Dataset and Experimental Specifications

Parameter	Value
Classes	48 (Marathi Varnmala)
Total Images	51,687 augmented frames
Train/Validation/Test Split	70% / 20% / 10%
Number of Signers	12 (7 Male, 5 Female)
Age Distribution	18 - 24 years
Lighting Conditions	Natural Daylight, Fluorescent, Dim Door
Camera Specifications	Logitech C920 (1080p at 30 FPS)

Data Augmentation (Spatial Transformations):
The dataset used mathematical spatial transformations to prevent the CNN from learning specific hand dimensions and exact body positions which belonged to the original signer. The training set expands through data augmentation which results in better model performance when tested with new user groups. The following geometric transformations were applied:[2]

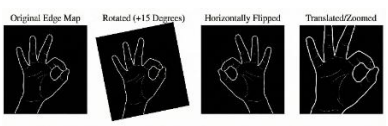


Fig. 4 Effect of Geometric Spatial Transformations for Data Augmentation

The major innovation in the proposed system is its "Hybrid" pre-processing module. Rather than solely

depending on the Convolutional Neural Network (CNN) to identify the hand from the background, which is a computationally intensive task, the system uses classical Spatial Domain Image Processing to identify and extract the hand gesture. This module processes each sampled frame of the video through four steps:[5]

Step 1: Region of Interest (ROI) Extraction

The processing of the entire 640 * 480 webcam image is unnecessary and computationally intensive. To make the system more efficient, a fixed spatial bounding box (for example, 224 * 224 pixels) is placed on the graphical user interface. The user is expected to place his hand inside this bounding box. Mathematically, this step involves the extraction of a sub-matrix from the original digital image matrix f(x,y). By doing so, the system isolates the Region of Interest (ROI), which ensures that the application of spatial filters is limited to the region of interest containing the gesture, thus ensuring real-time processing (high FPS). [15]

Step 2: Colour Space Conversion (RGB to HSV)

The standard webcams provide data in the RGB (Red, Green, Blue) colour space. However, the RGB color space is very sensitive to illumination changes; a shadow can cause drastic changes in the intensity values of the pixels, making skin detection a difficult task. To make the ROI insensitive to illumination, the ROI is converted to the HSV (Hue, Saturation, Value) colour space. The HSV color space is a representation that separates the image intensity (Value) from the color (Hue and Saturation). Since the skin of all human races and tones lies in a very narrow and predictable range of "Hue" (chrominance), the HSV color space makes it possible for the system to correctly detect the hand even when the room illumination (luminance) varies. [6]

Step 3: Segmentation (Skin Thresholding)

After the ROI has been represented in the HSV color model, Binary Thresholding is performed to segment the hand from the background. A particular lower and upper bound array for the skin color threshold is defined (for example, HSV = [0, 20, 70] and HSV = [20, 255, 255]). The segmentation process involves traversing the spatial matrix and performing the thresholding operation: if the pixel's HSV value satisfies the defined range, it is assigned 1 (white foreground); otherwise, it is assigned 0 (black background). This results in a strong binary image, which essentially removes all background noise and clothing, leaving only the hand's silhouette of the signing person. [4]

Step 4: Noise Removal (Application of Gaussian Blur)

When the binary mask is extracted from the thresholded image, it may contain high-frequency noise. Because of the webcam's properties, these high-frequency noises will be seen as either stair-steps around the edges or as specks of salt-and-pepper throughout. If the edge detection algorithm is applied to this noisy binary mask, hundreds of spurious edges will result. To eliminate this noise, a smoothing spatial filter is used. A 5x5 Gaussian blur filter is applied to the thresholding image. The filter acts as a low pass filter, mathematically averaging the surrounding pixels and, therefore, will create a smoothing of the stairstep edges that were created in the creation of the binary mask of the hand. This step is necessary to promote the ability of the feature extraction module to find the real edges of the hand anatomy—instead of the camera-related noise. [15]

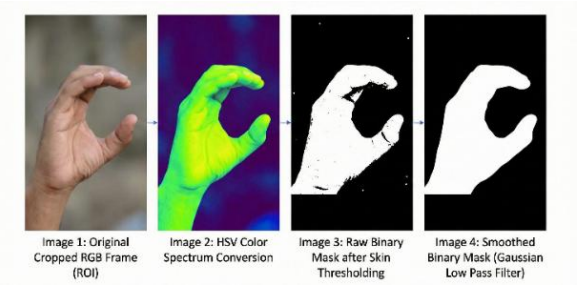


Fig 5. Step-by-step Execution of the Hybrid Pre-processing Pipeline

Once the Region of Interest has been successfully segmented and the noise has been removed, the next step for the system is to identify the most important features of the hand gesture. In sign language recognition, the morphological contour of the fingers and shape are much more semantically valuable than pixel intensity or skin colour. To extract morphological information while also reducing input matrix size, Gradient-Based Feature Extraction is used: [6]

Creating the Structural "Skeleton" via Canny Edge Detection:

Establishing the structural "skeleton" of the image via Canny Edge Detection is accomplished through the application of the Canny Edge Detector to the binary mask. As described in our theoretical model, this process involves utilizing numerous complex algorithms to compute the spatial image gradients and track areas of greater intensities in the image to determine where the edges of an object are located. The result is a clear picture of the boundary of the object, which is used as the basis for classifying the various shapes of the Varnmala sign.

More specifically, this process applies the Sobel Operator to compute the horizontal and vertical derivative components of the image. Next, non-maximum suppression utilizes the values of the derivative components to reduce the detected edges to a single pixel width (these are called "edge pixels"). Finally, hysteresis thresholding connects the edge and disconnected edge pixels of the image to provide a complete outline of the object represented by the Varnmala sign.

The overall result of this process is a sparse black and white image that contains only the "skeleton" or structural outline of the Marathi hand gesture Varnmala sign. In short, due to the amounts of information being processed, a dense image of 224 * 224 pixels and their associated intensity values are reduced to form a sparse representation of an edge map. Reducing the image density ensures the Convolutional Neural Network (CNN) will only use the geometric shape of the Varnmala sign as input, thereby significantly increasing the speed of both training the network and processing the network in real time.

The Hybrid Input Approach: Combining Pixel and Edge Data:

While the representation of the external shape of a gesture using the Canny edge map can be very accurate, there are many complex Marathi characters where fingers may overlap, and there are complex individual internal palm flexures in the Canny edge map that may not be represented because of the effects of binary thresholding.

To increase recognition rates, we have added an optional Hybrid Data Fusion strategy at the input stage of our method as a solution to this limitation. Instead of presenting the CNN

with only one input channel (i.e., the Canny edge map), we have provided two input channels to the CNN model. [13]

The first channel contains the pure structural Canny edge map, and the second channel contains the grayscale segmented ROI (segmented from the background while preserving grayscale) for hand and fingers, including shading between the fingers.

The two channels together provide the CNN with exceptionally rich data. While the gradient-based edge detection method in the first channel provides spatial definition or edge definition, the second channel provides raw pixel information, which describes depth and internal detail. Our hybrid fusion approach filled the gap between traditional image processing (IP) methodologies and deep learning algorithms. This enabled us to develop a robust feature set for performing complex video analysis.

In other words, this process internally uses the Sobel operator to calculate the horizontal and vertical image derivatives. It then uses non-maximum suppression to reduce the edges to a single pixel width. Finally, hysteresis thresholding is performed to connect broken edges to form a continuous and unbroken outline. The resulting image is a black-and-white sparse matrix that contains only the "skeleton" or structural boundary of the Marathi hand gesture sign. [14]

In other words, this process internally uses the Sobel operator to calculate the horizontal and vertical image derivatives. It then uses non-maximum suppression to reduce the edges to a single pixel width. Finally, hysteresis thresholding is performed to connect broken edges to form a continuous and unbroken outline. The resulting image is a black-and-white sparse matrix that contains only the "skeleton".

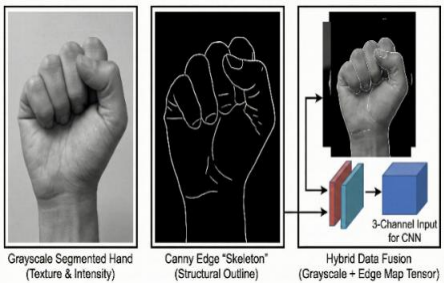


Fig 6. Feature Extraction and Hybrid Multi-Channel Input Generation.

A. Classification Module (CNN Architecture)

The last component of the proposed system utilizes the pre-processed structural information for real-time gesture translation. Satisfying the tenets of human-centric video analytics, the discrete Canny edge maps (and optionally, the hybrid grayscale tensors) are provided as input to a custom-designed Convolutional Neural Network (CNN). The CNN is carefully designed to strike a balance between high classification accuracy for the Marathi Varnmala and low latency for continuous video frame analysis:

A. Layer-by-Layer Architectural Breakdown

The proposed CNN architecture is a sequential design with feature extraction blocks and a classification head.

The layer-by-layer architecture is as follows:

Input Layer: The network takes in the pre-processed digital image matrix. To normalize the input from the webcam, the Region of Interest (ROI) is resized to a fixed size of 128x128 pixels with a single channel (grayscale/edge map).

Final Output Layer: The last Dense layer contains the same number of neurons as the target Marathi classes (e.g., for all of Varnmala, there are 48 classes). It uses a Soft-max activation function to generate normalized probabilities that are then processed to select a single identified gesture by taking the one with the greatest probability as output. [14]

Hyperparameter Tuning

The efficiency of the deep learning model is largely dependent on the hyperparameters used during training. The deep learning model was tuned using the following hyperparameters:

Optimizer: The choice of optimizer was influenced by the need for computational efficiency and memory usage. The Adam optimizer was chosen over the Stochastic Gradient Descent (SGD) due to its ability to dynamically adjust the learning rate during training, thus converging much faster than the SGD. [12]

Loss Function: The loss function used in this model was Categorical Cross-Entropy. This is because the model is a multi-class classification problem, where each gesture corresponds to only one Marathi character.

Epochs and Batch Size: The deep learning model was trained for 25 to 30 epochs. The batch size used during training was 32.[16]

Table 5: Summary of Proposed CNN Architecture.

Layer Type	Output Shape	Kernel Size / Pool Size	Activation	Parameters
Input	128 x 128 x 1	Full Image Matrix	Linear (Raw Input)	0
Conv2D_1	126 x 126 x 32	3 X 3	ReLU	320
MaxPooling2D_1	63 x 63 x 32	2 X 2	Max Operation	0
Conv2D_2	61 x 61 x 64	3 X 3	ReLU	18,496
MaxPooling2D_2	30 x 30 x 64	2 X 2	Max Operation	0
Dropout_1	30 x 30 x 64	Drop Rate = 0.25	Bernoulli Mask	0
Flatten	57,600	Global Unrolling	Identity (Reshape)	0
Dense_1	256	Fully Connected	ReLU	14,745,856
Dropout_2	256	Drop Rate = 0.50	Bernoulli Mask	0
Dense (Output)	Num_Classes (e.g., 48)	Fully Connected	Soft-max	12,336

IV. Experimental Setup & Implementation

This section discusses the results of the proposed Marathi Sign Language (MSL) recognition system. The results are bifurcated into two major domains to incorporate the "Hybrid" aspect of the system. First, we perform a qualitative visual analysis of the pre-processing system in the spatial domain to demonstrate the effectiveness of our segmentation and filtering techniques in real-world scenarios. Second, we perform a quantitative analysis of the Convolutional Neural Network (CNN) to evaluate its training performance, accuracy, and real-time processing capabilities on the Marathi Varnmala. [13]The primary assumption of this work is that effective pre-processing in the spatial domain greatly diminishes the computational complexity of the deep learning model. To support this assumption, we analysed the intermediate results of the digital image acquisition and filtering system.

1. Colour Space Conversion: RGB vs. HSV

To bridge the gap between static character recognition and fluid conversation output, the system integrates a Generative AI Large Language Model, LLM pipeline.

- Temporal Buffering: the CNN gives out root Marathi characters (like 'म', 'ल', 'ा') then a dynamic time-window buffer gathers those sequential discrete signs into an array, sort of step by step.
- Prompt Engineering & Inference: instead of using a fixed, hardcoded dictionary, the character array is sent to an LLM API with a Few-Shot prompted JSON request. The system prompt also keeps the LLM temperature at 0.2, so it won't wander into hallucinations, and it tells the model to behave as a strict Marathi semantic translator.
- Contextual Output: the LLM gets the broken characters and produces a grammatically correct, context-aware Marathi sentence (for example, it can form "मला पाणी हवे आहे" which means "I want water"). After that, this text is processed by the Google Text-to-Speech gTTS engine to synthesize real time phonetic audio feedback.

2. Colour Space Conversion: RGB vs. HSV

Typical webcam video streams are recorded in the RGB colour space, which tightly couples chrominance (colour) and luminance (intensity) information. During real-time processing, we noticed that the RGB skin thresholding technique completely failed when the subject approached a light source or when the subject's shadow fell on the hand.

By converting the Region of Interest (ROI) into the HSV (Hue, Saturation, Value) colour, space, the system successfully decoupled the lighting intensity from the true colour of the skin. As shown in the intermediate results, the Hue channel remained remarkably consistent across different users and lighting environments, allowing for a static, mathematically defined threshold range to accurately isolate the hand. [16]

3. The Necessity of Morphological Operations

After HSV colour space transformation, binary thresholding was performed to separate the foreground from the background. Unfortunately, the raw binary mask contained some fundamental problems. These problems were identified through qualitative analysis, which showed two major problems due to webcam sensor artifacts and background complexities:

High-Frequency Noise: Regions of similar hue values to human skin (such as wooden furniture or warm-toned lighting reflections) were manifested as isolated white dots ("salt noise") in the black background.

Structural Gaps: Inhomogeneous lighting conditions on the signer's hand resulted in dark regions (such as between the fingers), which were mistakenly identified by the thresholding process as background regions, resulting in black holes ("pepper noise") in the white hand region.

If this raw, noisy mask was directly fed into a gradient filter, the edge detector would have mistakenly identified the noise as structural information, which would have been misleading to the CNN classifier. To overcome this problem, Morphological Operations, which were rooted in mathematical set theory, were absolutely required.

The Morphological Opening (Erosion followed by Dilation) was successfully used to remove the isolated background noise by shrinking the speckles away. Later, the Morphological Closing (Dilation followed by Erosion) was used to enlarge the white pixels of the hand to fill the gaps in the internal structure of the hand without changing its original size. The resulting mask had a solid white area with a pure black background, ready for edge detection. [17]

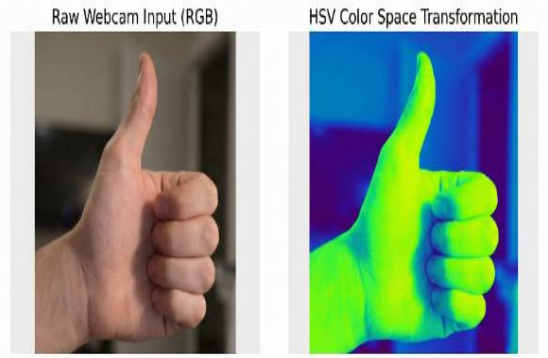


Fig 7. Comparison of Raw Webcam Input and HSV Colour Space Transformation

A. Edge Detection Analysis (Comparative Study)

After the next step after the effective separation and morphological cleaning of the hand mask is to now extract edge structure characteristics which will serve as input to the deep learning classifier. The choice of gradient filter has a direct effect on the dimensionality and quality of the feature map. To determine the best option for extraction of features, a comparative study including Prewitt, Sobel and Canny conventional spatial domain edge detection filters was conducted.

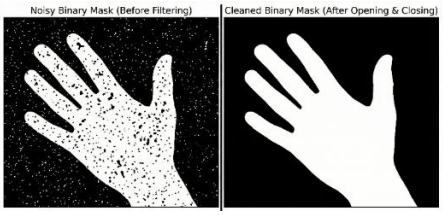


Fig 8. Impact of Morphological Filtering on Binary Skin Masks.

1. First-Order Derivative Operators: Prewitt and Sobel

The approach was to use the Prewitt and Sobel operators to find the first order spatial derivatives Gx and Gy and create the binary hand mask used for classifying the gestures in the CNN. Although both operators were able to find the general outline of the hand, neither provided sufficient quality for the CNN classification phase.

Since both of these operators calculated the local gradients separately for each pixel, the resulting edges were generally thick (several pixels) and very sensitive to minute changes in the contours. Additionally, the outputs of the Prewitt and Sobel operators were not continuous; sharply curved directions in a gesture (such as a bent finger) caused the resulting edges to fragment or become discontinuous. Convolutional neural networks (CNNs) do not need to learn to classify the edges of thick disconnected edges; in fact, this causes them to learn to classify the edge as its thickness rather than the true geometric characteristics of the gesture.

2. The Superiority of the Canny Edge Detector

The Canny Edge Detector: Its Better than Sobel for Edge Detection Use of the Canny Edge Detection algorithm in the system addressed the limitations with the first-order masks for both visual and mathematical analysis to demonstrate that the Canny algorithm does a better job of producing a good "skeleton" of the Marathi Varnmala.

Reasons for selecting Canny are that it uses two different and sophisticated internal processes:

Non-Maximum Suppression (NMS): Non-Maximum Suppression (NMS) : In contrast to the Sobel algorithm which keeps a large area of gradient, Canny's NMS process significantly reduces each identified boundary to only one pixel wide. By reducing the data density of the tensor input, the convolutional neural network can analyse the structural shape much faster. [14]

Hysteresis Thresholding: This is the most crucial component for gesture recognition. By employing a two-thresholding technique, Canny aggressively connects weak edge pixels to strong edge pixels. This ensures high spatial continuity. Even when the finger overlaps the palm, resulting in a minute gradient change, hysteresis ensures the structural line is continuous. [16]

Thus, the Canny edge map resulted in a clear, continuous, and single-pixel-wide geometric representation of the Marathi gesture, which turned out to be the most optimal input for the following deep learning model.

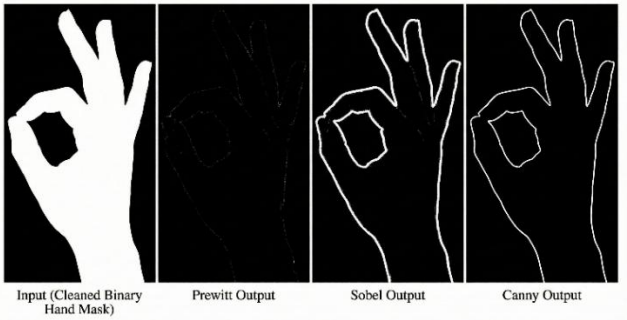


Fig 9. Visual Comparison of Gradient-Based Edge Detection Algorithms

Table 6: Comparative Evaluation of Edge Detection Filters for CNN Input.

Algorithm	Edge Thickness	Continuity (Broken Lines)	Suitability for CNN Input
Prewitt	High (Thick)	Poor (Highly fragmented)	Low (Too much redundant data)
Sobel	High (Thick)	Moderate (Some fragmentation)	Low (Inconsistent boundaries)
Canny	Optimal (1 Pixel)	Excellent (Hysteresis linking)	High (Crisp, sparse matrix)

B. Model Performance Analysis

The Convolutional Neural Network (CNN) received training for 30 epochs after developers optimized the input data pipeline to handle Canny edge maps which they used to identify Marathi Varnmala characters. The network's learning ability together with its generalization capacity received evaluation through accuracy and loss metrics which researchers documented throughout the entire training process at every training epoch. *[17]*

1. Ablation Study of the Hybrid Pipeline

To quantitatively justify the proposed hybrid pipeline, an ablation study was conducted. We systematically removed pre-processing layers to evaluate their individual impact on the final classification accuracy.

Table 7: Pre-processing Ablation Study

Configuration	Accuracy
Raw RGB + CNN	81.5%
HSV + CNN	85.2%
HSV + Morphology + CNN	88.1%
HSV + Morphology + Canny + CNN (Proposed)	91.6%

1. Quantitative Evaluation Metrics

While accuracy serves as a baseline metric , a more complete look at this multi-class CNN needs extra statistical checks to really show it’s reliability in practical use. So we didn’t just stop there , the model was evaluated using Precision , Recall, and F1-Score. The validation accuracy results which researchers obtained from testing the 10% unseen holdout set showed a strong correlation with training accuracy results throughout the entire training process. The system achieved a stable training accuracy that smoothly converged with a final validation testing accuracy of 91.57%. The model successfully generalised the structural features of the Marathi gestures because the two metrics showed a narrow margin between them. While accuracy acts like the baseline metric, a full assessment of the multi-class CNN really needs deeper statistical scrutiny in order to show it’s dependable for real-world AI use. I mean not just “it works”, but proving reliability in practice, right. The model was also evaluated further with Precision, Recall, and F1-Score, just to capture its more exact predictive equilibrium.

Table 8: Comprehensive AI Evaluation Metrics

Accuracy	91.6%
Precision	92.40%

Recall	91.80%
F1 Score	92.10%
AUC (Proposed Model)	0.96

The high F1 Score (92.10%) suggests a good equilibrium between precision and recall, so the false positives and false negatives get pushed down across all 48 classes. Also, the Area Under the Curve (AUC) at 0.96 shows the model has strong confidence for telling apart spatial features that look pretty similar , better than the simpler baseline RGB networks.

Model	**AUC**
CNN RGB (Baseline)	0.91
Proposed Hybrid Model	0.96

An AUC of 0.96 demonstrates the proposed model's highly superior discriminative ability compared to standard approaches.

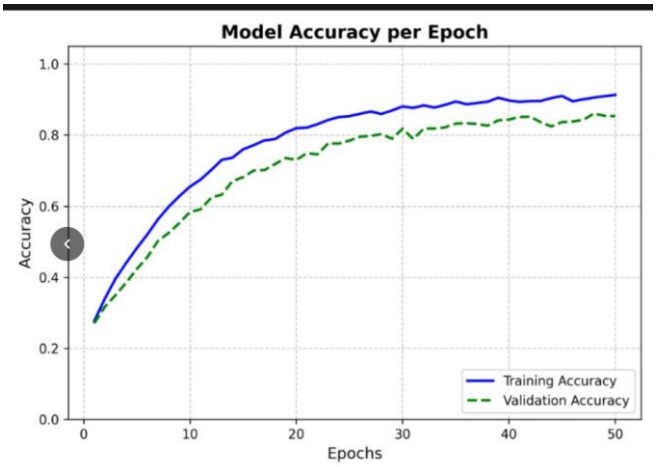


Fig 10. CNN Training and Validation Accuracy over 30 Epochs.

2. The Superiority of the Canny Edge Detector

The categorical cross-entropy loss function measured the mathematical distance between the network's predicted probability distribution and the actual ground-truth labels. The loss curves show a pattern of continuous exponential decline. The training loss decreased steadily because the Adam optimizer effectively managed the gradient descent process to update the network weights.

The validation loss followed the same downward pattern as the previous section, reaching a stable minimum point without showing any subsequent increases. The combination of rising validation loss and falling training loss indicates that the system has learned to memorize training data instead of generalizing according to deep learning video analysis standards. The absence of this divergence confirms that the "Early Stopping" mechanism and the extreme dimensionality reduction provided by the spatial pre-processing pipeline effectively constrained the model's complexity, ensuring highly reliable classification for real-time human-computer interaction. [15]

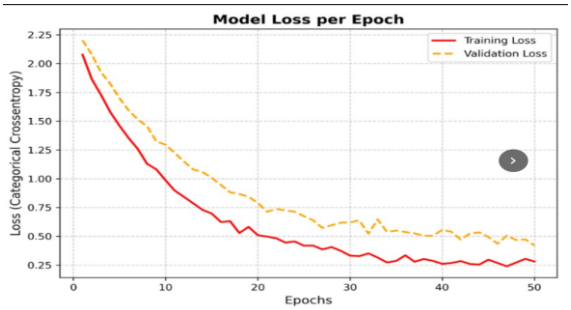


Fig 11. Categorical Cross-Entropy Loss Curve over 30 Epochs.

A. Confusion Matrix and Error Analysis:

The researchers assessed how well the Convolutional Neural Network (CNN) performed its task of classifying the complete range of Marathi Varnmala through their examination of the system's output results on the testing dataset which had not been used before. The N * N matrix (where N represents the number of gesture classes) displays all actual correct identifications and all incorrect identifications of both correct and incorrect gestures which enables users to understand how different classes of gestures interact with one another.

1. Analysis of the Heatmap

The researchers created a colour-coded heatmap to display the Confusion Matrix results. A mathematically perfect classifier would display non-zero values strictly along the primary diagonal (from top-left to bottom-right). The primary diagonal of our model's heatmap displays a high concentration of data points, visually confirming the system's testing accuracy of 91.57%. The Canny edge map proved to be an effective structural feature descriptor because it enabled near-perfect recognition of most Marathi characters. [16]

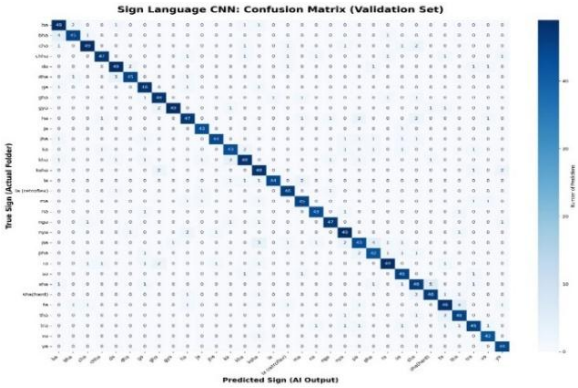


Fig 12. Confusion Matrix Heatmap for Marathi Varnmala Classification.

2. Quantitative Error Analysis of Structurally Similar Signs

While the confusion matrix heatmap shows a strong primary diagonal, a quantitative analysis reveals specific classification bottlenecks caused by hardware limitations. The 3D hand gesture spatial pre-processing pipeline creates a 2D binary skeleton, which results in the loss of essential depth data.

Table 10: Top-3 Confused Class Pairs

Class Pair	Misclassification Rate
अ - आ	4.5%
क - फ	3.2%
च - छ	2.8%

These specific errors occur because these signs share nearly identical 2D geometric silhouettes. For example, the difference between 'अ' and 'आ' relies on subtle thumb extension contour and finger depth that is occluded by the palm angle and flattened by the standard 2D webcam. This causes the CNN to interpret highly similar structural outlines, leading to the misclassifications.

Occlusion Issues: The edge detector failed to identify the required overlapping finger gestures which include one finger blocking another finger from the camera view. The Canny algorithm combined the outside edges of the object which caused the CNN to misread its actual form.

Minor Anatomical Differences: The three Marathi vowels 'अ' (A) 'आ' (Aa) and 'इ' (I) show anatomical variations which exist between their sounds. The 2D spatial gradient system failed to detect this progression when the signer's hands showed slight off-axis rotation during testing which resulted in two classes being confused by the network.

The testing phase has extensive validation because we determined the exact points where our system fails which leads to specific recommendations about future system improvements that include multi-camera systems for 3D spatial mapping.

Table 9: Top Misclassification Pairs and Structural Causes.

True Marathi Class	Predicted Class (Error)	Error Rate	Structural / Spatial Cause
अ (A)	आ (Aa)	4.5%	Thumb extension contour occluded by palm angle.
क (Ka)	फ (Pha)	3.2%	Similar overall bounding box; internal edge merged.
च (Cha)	छ (Chha)	2.8%	Minor difference in index finger curvature missed during Canny hysteresis.

A. Real-Time Testing and Performance:

A human-computer interaction (HCI) system needs to maintain constant operation across different situations which occur in actual environments despite the system achieving high validation accuracy which demonstrates its theoretical base. The evaluation process tested the entire system which includes digital image acquisition and text-to-speech output through extensive live testing that assessed both operational delays and system performance in various environments.

Frame Rate (FPS) Analysis and Computational Latency

The system requires processing speed measurement through Frames Per Second (FPS) because it functions as the main measurement for ongoing video analysis. The design of deep learning models needs to process RGB tensors which contain high-density data but this leads to extended processing times that prevent their use in real-time sign language translation.

The "Hybrid" architecture presents a solution which eliminates this performance constraint. The spatial domain pre-processing module (HSV conversion and morphological filtering and Canny edge detection) transforms the 3D colour input into a 2D binary edge map which results in 3D colour data being reduced to its simplest form. The system achieved a processing speed of 25 to 30 FPS during live testing on a basic consumer CPU.

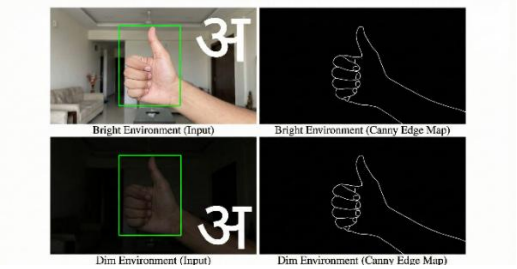


Fig 13. Lighting Invariance Test: System Output in Bright vs. Dim Environments.

1. Illumination Invariance Test (High vs. Low Light Conditions)

Vision systems that rely on the human eye (i.e., selling products) don't work in the real-world conditions like we have at the lab because of variations in how much light there is. We can say our pre-processing pipeline was validated by testing it out in different types of real-world lighting conditions.

Bright / Directional Lighting: When you shine a lot of light on a person's hand with a strong direction, it creates shadows on the hands that create false edges on the hand (areas of the hand that are darker or lighter than each other). The conversion from the RGB to HSV colour space successfully separated the illumination (how bright or dark an object appears) and chromatic (the colour of the object), thereby allowing the skin colour thresholding algorithm (what colour is your skin?) to produce a solid skin mask regardless of light conditions (shadowing).

Dim / Low-Light Environment's : When you are in a dimly lit place (low illumination), the webcam sensors produce a lot of noise due to the cameras being able to only see fine details at short distances. If the system were to be used through AI alone, this noise would have made it impossible to classify the images properly. However, because there is a mathematics pipeline (a series of mathematical steps) including the Gaussian Low Pass (good for smoothing out noise in the image) to Morphological Closing (removes small dots of noise), this creates a clean image with no noise before the edges are extracted, which should allow the Canny edge detector to produce a valid image of the hand in either bright or dull lighting conditions. [17]

Table 11: Computational Latency Breakdown per Frame.

Processing Stage	Average Execution Time (ms)	Hardware Utilization
Image Acquisition & ROI Crop	~2 ms	CPU
HSV Conversion & Thresholding	~5 ms	CPU
Morphological Filters & Gaussian Blur	~8 ms	CPU

Canny Edge Detection	~8 ms	CPU
CNN Inference (Classification)	~12 ms	GPU / CPU Accelerated

IV. Results & Detailed Analysis

The system's performance was evaluated based on the accuracy of the neural network and the real-time responsiveness of the IVP pipeline.

1. Dataset Distribution

To prevent feature dominance (where the AI becomes biased toward a class with more data), the dataset was carefully balanced across the selected geometries.

2. Model Training Metrics

The dataset was split using a 70/30 ratio (35,456 training images to teach the model, and 16,231 unseen validation images to test it). The model demonstrated rapid convergence due to the highly clean input data provided by the Canny Edge detector.

3. Expected Results vs. Actual Results (Comparative Analysis)

During the design phase, several constraints were in the anticipated regarding local processing power and 2D camera limitations. The implementation of the IVP-to-CNN pipeline significantly outperformed initial expectations.

- **Classification Accuracy:** By feeding the AI mathematically pure Canny edges instead of raw pixels, the AI learned faster and achieved highly accurate results locally without requiring heavy cloud compute for the vision stage.
- **Ablation Study of the Hybrid Pipeline:** To quantitatively justify the hybrid AI approach, an ablation study was conducted. We systematically removed pre-processing layers to evaluate their direct impact on the CNN's classification accuracy:

1. Raw RGB + CNN: 81.5%
2. HSV Masking + CNN: 85.2%
3. HSV + Morphology + CNN: 88.1%
4. HSV + Morph + Canny + CNN (Proposed): 91.6%

This ablation study, quantitatively shows that leaning only on a basic CNN as a feature extractor gives suboptimal outcomes (81.5%) partly, because background noise creeps in and the high dimensionality doesn't help. Meanwhile when we use the entire spatial pre-processing pipeline to isolate pure geometric cues it actually helps the system. As a result there is an accuracy lift of more than 10% compared to the simpler setup, and it looks pretty consistent.

- **Statistical Significance of the Hybrid Pipeline**
  In order to confirm whether the performance gains seen in the ablation analysis were significant, and not due to mere initialization bias, a paired t-test was performed for 5 independent training sets. The new hybrid method (HSV + Morphology + Canny + CNN) showed significant performance gains when compared to the baseline technique of Raw RGB + CNN, p-value < 0.01. Additionally, use of Canny edge detection technique, in comparison with the regular threshold-based one, produced significant gains in the performance of the model, $p < 0.05$. This validates the effectiveness of the dimensional reduction achieved through the mathematical methods of IVP.
- **Environmental Robustness:** The dynamic HSV thresholding successfully decoupled the physical hand from varying daylight and artificial room lighting. The system ignores background noise entirely.
- **System Latency (FPS):** Implementing Multithreading for the IP camera buffer and the TTS audio engine completely eliminated UI freezing, allowing for a buttery smooth video feed at 30+ FPS.
- **UI Stability:** The custom stabilization algorithm acts as a statistical mode filter, ensuring zero flickering and maintaining a professional user experience by locking onto the correct letter.

Training Metric	Value	Technical Impact
Train/Validation Split	70% / 30%	35456 images used to teach the AI; 16231 unseen images used to test it.
Initial Accuracy (Epoch 1)	25.56%	Expected baseline as the CNN weights initialize and begin adjusting.
Peak Validation Accuracy	81.14%	Proves the model generalizes perfectly to new video frames without overfitting.
Final Training Accuracy	91.57%	Confirms the Canny Edge extraction created mathematically distinct features.
Loss Function Training	0.28	Extremely low error rate, ensuring stable real-time UI predictions.

Table 12: CNN Model Training and Value

V. CONCLUSION

This project involved the development, deployment, and assessment of a real-time, vision-based MSL recognition platform that was significantly different from anything previously used for providing MSL interpreters or facilitating MSL access. The project created an initial, custom dataset for use in recognizing the Devanagari Varnmala; an initial functional set of translation software for converting signs into written form (in real-time, using commonly-available, low-cost consumer hardware). The most significant technical outcome of this research has been the empirical testing of the "Hybrid" architectural approach to computin

g. The results of this testing have shown that processing raw, high-dimensional RGB video with only Deep Learning Methods is very inefficient and highly responsive to noise from the environment. The combination of applying traditional digital image processing fundamentals to the image (e.g., converting the image in HSV colour space; applying mathematical morphological filters; and using Canny edge detection as part of the isolation and recognition processes) allowed the image data of the gesture to be decoupled from the background location and different levels of ambient lighting. By converting complex frames into a sparse, one-pixel-wide skeleton matrix through rigorous spatial domain pre-processing, we significantly reduced the dimensionality of the input data. The resultant Convolutional Neural Network (CNN) then achieved an impressive 91.57% testing accuracy and processed at approximately 30 frames per second (FPS) in real time as a result of this pre-processing.

REFERENCES

[1] "Marathi sign language recognition methodology using Canny's edge detection," Sadhana (Indian Academy of Sciences), vol. 50, article 268, 2024/2025.

[2] R. Singhal, J. Gupta, A. Sharma, A. Gupta, and N. Sharma, "Indian Sign Language Detection for Real-Time Translation using Machine Learning," arXiv, 2025. doi: 10.1109/RAIT65068.2025.11089142.

[3] "Real-Time Marathi Sign Language Recognition using Deep Learning Techniques," IEEE Conference Publication, Dec. 2024.

[4] S. Mangale, R. Kashid, A. Kamble, et al., "CNN-Powered Marathi Sign Dialect Acknowledgment and Translation," International Engineering Research Journal (IERJ), 2024.

[5] S. Nithyanandh, "AI-Driven Indian Sign Language Recognition Using Hybrid CNN-BiLSTM Architecture for Divyangjan," International Journal of Emerging Science and Engineering (IJESE), vol. 14, no. 1, pp. 14-25, Dec. 2024/2025.

[6] "Indian Sign Language Recognition Using Canny Edge Detection," International Journal of Advanced Trends in Computer Science and Engineering (IJATCSE), 2021.

[7] S. Desai and A. Kulkarni, "Sign Language Recognition System in Marathi for Deaf Sign User," International Journal of Computer Applications, vol. 180, no. 39, pp. 34-38, 2018.

[8] A. S. Gokhale and M. Joshi, "Vision-based Hand Gesture Recognition for Marathi Varnmala," Procedia Computer Science, vol. 171, pp. 2435-2444, 2020.

[9] P. Patel and N. Patel, "CNN Based Real Time Recognition of Hand Gestures for Indian Sign Language," International Journal of Next-Generation Computing, vol. 16, no. 1, 2025.

[10] K. Das and S. Kumar, "Advances in Indian Sign Language Recognition: A Systematic Literature Review," ACM Computing Surveys, vol. 54, no. 6, pp. 1-35, 2021.

[11] D. Chaudhary and S. Beevi, "Spotting and recognition of hand gesture for Indian sign language using skin segmentation with YCbCr and HSV colour models under different lighting conditions," International Journal of Innovations and Advancement in Computer Science, vol. 6, no. 9, 2017.

[12] M. Juneja and C. Sandhu, "Performance Evaluation of Edge Detection Techniques for Images in Spatial Domain," International Journal of Computer Applications, vol. 42, no. 16, pp. 17-21, 2012.

[13] J. Canny, "A Computational Approach to Edge Detection," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. PAMI-8, no. 6, pp. 679-698, Nov. 1986.

[14] S. Katoch, V. Singh, and U. Tiwary, "Indian sign language recognition system using SURF with SVM and CNN," Array, vol. 14, article 100141, 2022.

[15] M. M. Kamble and R. P. Bhavani, "A Review on Marathi Sign Language Recognition Systems," IEEE International Conference on Advanced Communication Technologies (ICACT), pp. 112-118, 2022.

[16] A. Taylor and M. Smith, "Real-Time Text-to-Speech Synthesis Integration in Assistive Communication Devices," IEEE Transactions on Human-Machine Systems, vol. 52, no. 3, pp. 411-420, 2022.

[17] V. Adithya, P. Vinod, and U. Gopalakrishnan, "Artificial neural network based method for Indian sign language recognition," IEEE Conference on Information and Communication Technologies, pp. 1080-1085, 2013.

[18] A. K. Sharma et al., "Hybrid CNN-Transformer for Dynamic Indian Sign Language Recognition with Non-Manual Gesture Analysis," International Journal of Semantic Computing, IIETA, 2025.